

INTEGRATED DESIGN PROJECT 1
Final Report

Dementia Assistive Robot

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AI Tool Used	Purpose	Section(s) of Report OR Work Affected
Gemini	Generates conversational responses	Section 3.2, 3.5
Whisper	Converts recorded user speech into text for processing to be done by Gemini	Section 3.2, 3.4
Piper TTS	Converts generated text responses into verbal speech	Section 3.2, 3.4
Claude	Suggestions of proposed solutions to be integrated	All
ChatGPT	Report organisation, grammar and refinement of wording	All
Consensus	Assisted with literature review synthesis	Section 2

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1 Introduction

The world is slowly shifting towards an increasingly elderly population, enabling safe and independent living in the home has become an urgent public health priority. Malaysia is affected at a more accelerated pace compared to Western nations. According to Department of Statistics Malaysia (DOSM), Malaysia surpassed the threshold for an “ageing nation” with predicted population aged 65 and above hitting 8.0% of total population by 2026 [1]. One of the greatest threats to autonomy amongst the elderly is unintentional falling within their own household. This often leads to severe injuries and loss of personal independence. Traditional assistive technologies rely on passive monitoring and wearables emergency buttons. These solutions often result in poor user compliance, delayed emergency responses, and the inability to adapt immediately to the users environment. To address these problems, this project introduces, “PIPER”, a dementia assistive robot specifically engineered to bridge the gap between active home safety, health monitoring, and daily social companionship. PIPER operates as an intelligent in-home assistant that actively safeguards the user while offering intuitive, hands-free support.

1.1 Problem Statement

Elderly individuals, particularly those living alone, often face challenges in managing medication schedules, remembering daily activities, obtaining assistance during emergencies, and maintaining regular social interaction [2]. Existing solutions such as smartphone applications and emergency devices typically address only specific needs and may be difficult for elderly users with limited technical literacy to operate [3],[4]. Therefore, there is a need for an intelligent and user-friendly companion system that integrates voice interaction, emergency assistance, smart home control and mobility support to enhance elderly safety, convenience, and independent living.

1.2 Aim and Objectives

To address these gaps, this project is governed by the following core objectives:

1. To develop an AI-based voice interaction system capable of providing voice responses and answering simple queries with an average response time below 3 seconds.
2. To develop an emergency alert system capable of notifying caregivers through wireless communication (WhatsApp or Telegram) within 10 seconds of activation via fall detection.
3. To integrate voice-controlled smart home functionality capable of controlling selected household appliances such as lighting and fans with a command execution accuracy of at least 85%.

1.3 Scope of the Project

This section defines the precise operational boundaries of PIPER, outlining functions within its scope.

Physical architecture	CAD design and 3D printed with a multi-tiered base for component housing and low centre gravity weight distribution.
Core subsystems	• Mmwave radar sensor based fall detection • AI voice interaction with memory based • Automated emergency notification protocol
Target environment	Idle on a table or floor of a room the elderly uses most frequent.

Initially our robot was set to be mobile around one floor of the house. However, that has been proven to be too extreme due to the lack of budget and high-power consumption. Apart from that, another exclusion would be for obstacle removal around the house as it could potentially be a tripping hazard on its own and it would be using too much energy that isn't available to be used for other more important features.

1.4 Societal, Safety and Sustainability Context

1.4.1 Societal

PIPER directly addresses the global demographic shift toward aging populations by empowering elderly individuals to age in place safely. By reducing unintentional household falling by providing real-time fall detection. Thus, PIPER alleviates caregiver strain, reduces reliance on institutional care facilities, and helps older adults maintain personal autonomy and dignity.

1.4.2 Safety

Unintentional falls are a primary cause of traumatic injury and loss of independence among older adults. PIPER directly mitigates health risks through proactive hazard clearance and rapid post-fall alert escalation. To guarantee user safety during operation, the robot's physical body features rounded geometries to eliminate sharp impact edges.

1.4.3 Sustainability

To minimize lifecycle environmental impact, PIPER's structural frame and outer panels are manufactured using biodegradable PolyLactic Acid (PLA) sourced from renewable plant starches.

Additionally, the robot utilizes low-power embedded microcontrollers to optimize energy consumption during continuous monitoring.

1.5 Report Organisation

The report is organised into five sections. Section 1 introduces the project background, problem statement, aim and objectives, scope, societal safety and sustainability considerations of our robot. Section 2 presents the literature review covering different themes and features of the robot., before identifying the research gap. Section 3 describes the proposed methodology and design of Piper. Section 4 presents the project management part including its tools. Finally, section 5 concludes the report by summarising the proposed project, expected contributions, limitations and future improvements.

2 Literature Review

Rapid advancements in assistive robotics, computer vision, and artificial intelligence have driven significant research into technology-enabled elder care. As global demographics shift toward an aging population, maintaining physical safety and independence in domestic environments has become a critical objective. This chapter presents a critical synthesis of existing literature related to companion robots and active safety systems. The objective of this review is to evaluate the current state of technology in elderly care robotics, identify key functional limitations in existing commercial and academic platforms, and establish the theoretical foundation for PIPER. Rather than examining these technologies in isolation, this review analyses how social interaction, physical safety, and computer vision intersect in assistive home robotics.

2.1 Companion & Socially Assistive Robots

Animal Mimetic Bio-Robots are “pet like” interactive companion robot that are implemented for social, sensory and emotional therapy. Such “animal-like” robot are programmed to display “emotions and social expression” when interacts with human users. Systematic meta-analyses studies done by Lu, et al. [3] have demonstrated that pet type robots are effective at lowering neuropsychiatric symptoms like anxiety and depression in older adults with mild to severe dementia. One drawback of this robot is that they are prohibitively expensive and may lead to privacy protection issue.

Assistive Technologies like smartphone apps or wearables like biometric trackers are designed to track heart rate, fall detections and give reminders to users for timely meal and medications [3], [4]. They also aid healthcare workers by alerting them in the event of a heart attack or a fall. One of the drawbacks is that senior citizens may remove wearables as they might feel irritable for wearing items on their body every day.

Current assistive technologies for the elderly forces a compromise between emotional companionship and functional medical monitoring. Despite companion robots offer “emotional” presence but they still lack actual human interaction and are high-cost utilities. Conversely, standard digital solutions such as mobile applications and wearable lack cognitive interaction and a lack of physical embodiment causing the psychological impacts of loneliness.

Furthermore, there is a distinct lack of affordable, context aware assistive devices optimized for ensuring data privacy that can operate reliably within the constraints of modern independent living spaces.

2.2 Human Robot Interaction

Human–robot interaction is important in healthcare setting to providing companionship, reminders and assistance with daily activities. A mixed method of systemic review comprising of 86 studies found that social assistive robots reduce the workload of healthcare workers and slightly the emotions of elderly patients [5].

Recent research also shows that older adults do not accept all forms of interaction equally. In a year-long co-design project involving 28 senior citizens, participants were more accepting of transactional functions such as reminders and scheduling but were less comfortable with sharing

sensitive information and being monitored. Concerns included surveillance, privacy, autonomy, and the naturalness of interaction [6].

Regarding reminders on medications, Su, et al. [7] developed a conversation-based medication-management system using a companion robot and cloud. Their evaluation included 15 robot-end participants and the system achieved a reported System Usability Scale score of 81.333 %, indicating positive usability [7]. This is relevant to the proposed use of Piper and Gemini: natural-language interaction can make medication and daily routine support more accessible than a screen-based interface like a phone.

Overall, the strongest design implications for the feature in this project are to provide simple voice commands, explicit confirmations and useful transactional functions such as reminders and smart-home control. The evidence also indicates that acceptance cannot be assumed from technical capability alone but rather be evaluated with representative users for trust, perceived usefulness, ease of interaction, speech-recognition accuracy and willingness to use the system over time. Existing studies support the design direction but they do not directly validate the exact combination of LD2450 sensing, Piper speech, Gemini responses, and WhatsApp caregiver alerts.

2.3 AI-Based Voice Interaction

To implement the AI-based voice functionality we need a 3-stage pipeline which consists of speech-to-text (STT), large language models (LLM) and text-to-speech.

2.3.1 Speech-to-Text (STT)

Before the robot can perform anything, it must first be able to take the speech signal from the user and accurately transcribe the speech. Generally, how they do it in modern robotics is that they convert the physical audio signal of a user voice into digital text by typically routing the audio to a highly powerful cloud-based Speech-to-Text (STT) services rather than processing it with the AI robot itself [8],[9],[10]. The table below shows the general comparison for STT integration in voice AI robotics:

Table 2.3.1: Speech-to-text literature review summary

System	Strategy on Hardware	Performance and Settings	Problems encountered and Solutions
Microsoft Azure Speech-to-Text	Speech-to-Text features combined with Voice Activity (VAD) mechanism on Pepper robot	90.62% accuracy in controlled environments; uses user-calibrated silence	Problems: Fan noise and slow speech rhythms Solution:

		thresholds of roughly 1.8 s [8]	External USB microphone array and adjustable silence pauses
Google Cloud Speech-to-Text	Real-time speech recognition integrated into Furhat humanoid robot	Generally high accuracy, but it falls away noticeably on Swedish names and dialectal terms [9]	Problems: Stale audio buffers and premature user cut-offs Solution: Increased silence timeout (1.8 s) and red LED turn-taking indicators
OpenAI Whisper (Offline),	Post-session audio transcription and diarization, used to verify the quality of experimental dialogue	Highest baseline accuracy for offline processing of multilingual recordings [8], [9]	Problems: Computational latency too high for live use Solution: Restricted to post-interaction verification rather than live conversation
USB Sound Card & Lapel Setup	External USB sound card which is connected to a Raspberry Pi 4 pet robot	Reliable, low-cost analogue-to-digital conversion running from a standard power bank [10]	Problems: The Raspberry Pi has no integrated sound input Solution: Inexpensive external sound card paired with

			a compact lapel or headset microphone
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Cloud STT systems run into practical problems once they are placed inside a physical robot, as the hum of cooling fans and motors sits close to the microphone and can potentially mask the user's voice entirely, which designers typically address by moving an external microphone array nearer to the speaker.

A less obvious problem is turn-taking. Since standard speech systems use tight silence thresholds which is usually between 800 ms and 1.2 s and these regularly cut off older adults' mid-sentence, producing interruption rates above 40% [8], [9]. However, widening the listening window to around 1.8s gave older users room to speak at their own pace and brought interruptions down to 13.1% [8], or a mean of 3.18 times per session [9].

Whisper is currently used mainly to verify transcripts after the fact rather than during the conversation [8], [11], though the same authors point towards faster variants such as Whisper JAX as a route to running these models live.

2.3.2 Large Language Models (LLMs)

After the robot got the text from STT, it needs a brain to process the text and produce a reply. In terms of the research we have done, most robots project have moved away from rigid pre-written scripts and are now mostly using Large Language Models (LLM) which are more capable of simulating a real response that matches an actual person and capable of doing so with almost any subject [8], [9], [11].

Table 2.3.2: Large Language Model (LLM) literature review summary

Model	Dialogue and memory strategy	Conversational Advantages	Main Risks and Safeguards
Pinto-Bernal et al. (ChatGPT 3.5) [8]	Uses a layered approach that combines a predefined robot persona, a long-term	Provides greater flexibility in open-ended conversations and helps create a stronger sense of	Risk: Limited token space and frequent AI disclaimers can

	user profile, and recent conversation history.	continuity and social awareness between sessions.	interrupt the conversation. Safeguard: Summarising important information during the session and using clear anti-disclosure instructions.
Irfan et al. (GPT-3.5) [9]	Uses an empathetic persona built around the user's family, memories, and personal experiences.	Encourages greater user engagement, making the interaction feel more natural and more like talking to a familiar person rather than a device.	Risk: The model may provide inaccurate information or gradually lose its intended persona. Safeguard: Short persona reminders are included throughout the conversation to help maintain consistent behaviour.
Lima et al. (GPT-4o) [11]	Uses structured prompts that provide encouragement, praise, and motivational feedback during cognitive tasks.	Supports coherent conversations in both English and Spanish while adapting responses to the user's individual interests.	Risk: The model may produce responses that are off-topic or not sufficiently controlled. Safeguard:

			An AI-based moderator filters responses before they are delivered to the user.
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What makes an LLM useful in this setting is its ability to maintain a sense of continuity across different conversations through a layered memory system. At the end of a conversation, the model can be prompted to remember important personal details, such as a user’s hobbies, family, or upcoming travel plans, and store them securely in a long-term profile. When the user returns, these details can be brought back into the conversation, allowing the model to respond in a way that feels more familiar and attentive [8], [9], [11]. This makes the user feel more comfortable in a way that the robot actually understands them.

However, this approach also comes with several risks in which for example hallucination, where an LLM may confidently provide information that is incorrect or outdated, such as giving the wrong bus timetable. This can be especially confusing for older adults who may have difficulty checking whether the information is accurate [9], [11]. Another issue is persona drift, where the model gradually loses its intended conversational personality and starts giving generic responses such as *“As an AI, I do not have feelings.”* This can interrupt the sense of connection that the system is designed to create [8], [9]. A common way to reduce this problem is to include a short reminder of the model’s intended persona with each user message, rather than relying only on the original system prompt to maintain its behaviour throughout the conversation [8], [9].

2.3.3 Text-to-Speech (TTS)

The final stage of the pipeline is the Text-to-Speech. This is where the LLM's text output is turned back into audio. In early days of social robots, they use local, rather more flat sounding voices that often times disengaged the users quickly but nowadays, a lot of the algorithms and systems rely on neural Text-to-Speech (TTS) engines more which tends to carry more warmth, is more expressive and speaks at a natural pace [8], [10], [11].

Table 2.3.3: Speech-to-text literature review summary

Voice Engine	Voice Persona and settings	Emotional and cultural features	Pacing and latency mitigation
ElevenLabs [8]	Uses a high-fidelity cloud voice selected to match the user's preferred language and accent.	Warm and expressive voice that adds emotional depth and supports regional accents, including Flemish Dutch.	Streams sentences as they are generated, helping responses start quickly and keeping conversations smooth.
Unreal Speech API [11]	Pre-recorded female voice with adjustable speed and pitch.	Clear and easy to understand, with high usability ratings from older adults.	Slows speech to about 80% of the normal speaking rate to improve comprehension.
Amazon Polly (“Matthew Neural”) [9]	Natural-sounding male voice hosted in the cloud.	Friendly and engaging voice that matches the robot's visual appearance for a more relatable interaction.	Uses brief pauses and simple non-verbal cues to hide the typical 2–3 second processing delay.

A warm and natural-sounding voice can help build trust, but it is also important for the robot to sound familiar to the user. One way of achieving this is through linguistic mirroring, where the system adapts its speech to match the user’s language and accent. For example, Dutch-speaking older adults reported feeling more comfortable and safer when the robot used a regional Flemish Dutch accent rather than standard Dutch [8]. Similarly, older adults with hearing or cognitive difficulties may find normal speech rates too fast. Reducing neural TTS speech to around 80% of the normal rate has been shown to make the conversation easier to follow and understand [8], [11].

However, latency is still one of the main challenges in cloud-based voice systems. Communication with cloud services can introduce delays of around seven to nine seconds [8], [9], which can make the interaction feel unnatural. A useful solution is incremental streaming, where the system begins playing the response as soon as the first sentence is ready instead of waiting for the entire response to be generated. The remaining delay can then be made less noticeable through simple behavioural cues. For example, the robot could use blue eye LEDs to show that it is listening or briefly look away to give the impression that it is thinking about its next response [8], [11].

2.4 Fall Detection and Emergency Alert Systems

Fall detection systems across four articles were compared and the main differences are sensing modality, how the robot responds after detection, and how far each system has been validated.

Table 2.4: Literature review summary on fall detection and emergency alert systems

Title	Year	Methodology	Performance	Strengths	Weaknesses
Patrol Robot (APR) [12]	2019	YOLOv3 & SVM classifier on RGB images	Precision 100%, Recall 99.74%	Distinguishes resting from falls; relocates if uncertain.	Misses active falling; limited to post-fall detection.
EM-Fall [13]	2026	mmWave radar on humanoid robot	Robust across 8 indoor environments	Day/night operation; privacy-preserving; handles occlusion	Rarely explored; complex interference management
Wi-Fi Companion Robot [14]	2024	CSI disruption analysis via deep learning	Effective in real-world experiments	Non-line-of-sight; privacy-preserving	Requires Wi-Fi infrastructure; limited range
ElderCare Robot [15]	2024	YOLOv7 + RRT* path planning	96% fall detection accuracy	Continuous tracking; real-time posture detection	Relies on camera; potential illumination issues

Two fall detection systems use camera, Patrol Robot (APR) [12] and ElderCare Robot [15], with their performance obtaining the highest fall detection accuracies. Yet camera-based systems remain vulnerable to lighting, occlusion, and privacy constraints, which explains the main reason of EM-Fall [13] and the Wi-Fi Companion Robot [14] to shift towards using non-camera sensing approaches. After detection, the Wi-Fi companion Robot rescues the human, handles obstacles and contacts caregiver [14], while APR and ElderCare focus more on detection and navigation, and EM-Fall emphasizes continuous monitoring robustness across rooms [13].

2.5 Privacy and Ethical Considerations of Companion Robots

Privacy and ethical considerations are major concerns in the use of assistive robots for elderly and dementia care. Existing literature highlights concerns related to privacy, autonomy, dignity, consent, surveillance, and accountability in table 2.5.

Table 2.5: Literature review privacy and ethical considerations

Article	Focus	Key Findings	Implications
B. Deusdad [16]	Dementia-specific ethics	In dementia care, the familiar concerns remain privacy loss, loss of	Companion robots may raise fewer data-privacy concerns in some cases, yet concerns

		control, liberty restriction, deception, infantilization, and accountability, but dignity and acceptance become especially acute .	remain about misrecognizing machines as living beings and about replacing human caregivers .
C. Johnston [17]	Robotic care in homes	Home robots interfere with privacy by recording movement and sometimes visitors; they may also suppress autonomy in the name of safety	Ethics-by-design and ethics committees are proposed, but informed consent can be fragile because of coercion, complexity, and cognitive decline
C. Lutz et al. [18]	Robot privacy paradox	Privacy concerns reduce use intention, but perceived benefits can override them; respondents were most concerned about manufacturer-side data protection	Authors recommend investment in privacy-sensitive robotics and transparent communication about protections
Chorong Park et al. [19]	Elderlies' disclosure and attachment to robot pets	Pet-like design increased anthropomorphism and affection, while concerns about privacy, autonomy, dignity, and future harms persisted; the paper frames four tensions around consent, surveillance, data minimization, and human-care substitution	Suggests companionship should be paired with transparency, meaningful control, and privacy-preserving defaults.

Assistive robots operating in homes may collect data by recording elderlies' movements, interfering with privacy and autonomy [18]. Due to privacy concerns, elderlies are less willing to accept assistive companion robots. However, perceived benefits of robotic assistance may outweigh these privacy concerns. Ethical concerns of robots that intervene in users' activities for safety purposes may unintentionally restrict their dignity and autonomy. Based on the studies, anthropomorphic designs encourage emotional attachment to companion robots which also raises other concerns on substitution of human care [19]. These assistive robots should act as an alternative solution to support rather than replace human-caregivers. The literature review highlights the need for privacy-sensitive and ethical approaches in design of the dementia assistive robot, and these findings support the design of Piper as a privacy-preserving companion robot that avoids camera-based monitoring while providing care to elderlies.

2.6 Comparison of Existing Assistive Robots

The features of the top 5 assistive care robots were compared in Table 2.7. The closest robot that matches our criteria is Enabot EBO X, however, there are none that encompasses all features. Therefore, Piper, our dementia assistive robot, is proposed to fill these limitations.

Table 2.6: Comparison of existing assistive robots

Features / Robots	ElliQ [20]	EMIR (Emotional Intelligent Robot) [21]	Pebble – The Fall Detection Companion Robot [22]	Enabot EBO X [23]	Pepper [24]
AI voice interaction	✓	✓	×	✓	✓
Memory conversation	✓	✓	×	×	×
Fall detection emergency system	×	×	✓	✓	×
Smart Home Control	×	×	×	✓	×
Privacy-preserving (no camera)	×	×	✓	×	×

2.7 Research Gap

Overall, the literature review shows that existing robots provide individual or limited functionality such as AI-based interaction, fall detection, emergency assistance and smart-home support. The comparison of existing assistive robots in Table 2.6 proves the need for an assistive robot that provides the complete combination of features integrated into a single system, while maintaining privacy and memory in its database for dementia support.

Table 2.7: Research Gap of reviewed articles

Research Gap	Limitation	Proposed Solution in Piper
Lack of integrated assistive functionality	Existing robots provide different combinations of assistance and companionship but none integrates all identified requirements.	Integrate AI voice interaction, memory conversation, fall detection, automated emergency alert, smart home control into one platform.
Limited privacy-preserving monitoring	Uses camera-based monitoring that creates privacy concerns of being ‘watched’.	Use non-camera based sensing approach through mmwave radar sensor for fall detection.
Limited dementia-oriented interaction	The companion robots with many features are not specifically designed for dementia support.	Voice-based interaction and memory supported conversation while providing

		reminders designed for dementia elderlies.
Separation between safety and companionship	Fall detection and emergency systems are often separate from social functions.	Combine safety monitoring with AI companionship.
Limited consideration of accessibility	Some elderlies do not prefer wearables and mobile devices would require active user operation.	Provide hands-free voice interaction and automated emergency alert system.

3 Methodology

The methodology describes the process to develop Piper, our dementia assistive robot. This section covers the overall system architecture, hardware and software development, component selection, circuit and power considerations, and physical prototype design, safety, ethical, and sustainable design considerations. Engineering constraints and decision matrices are also considered and carried out before the selection of components and approaches.

3.1 Proposed Approach and Justification

In our project, we will be building a dementia assistive robot with the combined features of emergency fall detection, voice-based interaction and smart home control. This approach is an improved version compared to previous research on assistive robots and smart home environment. Our robot addresses the research gap in section 2.7 by applying the same general principle on a smaller scale by integrating fall sensing, artificial intelligence, and home automation into one platform while ensuring privacy and memory retention of LLM.

3.1.1 Engineering Constraints and Trade Offs

There are several factors that needs to be considered when implementing multiple features in a robot. For example, cost, durability, reliability and system complexity. To solve this issue and reach an optimised decision, multiple decision matrices table have been created to ease our selection for each feature.

In the decision matrices from tables 3.1.1a to 3.1.1e, different weightings are assigned to each criterion according to their importance. Each option is rated on a scale of 1 to 5, with 5 being the highest suitability. The weighted scores are calculated according to the formula:

$$Total\ score = \sum (weight \times rating)$$

The selected options are based on the highest total weighted score.

Table 3.1.1a: Decision Matrix for Types of Sensors for Fall Detection

Criteria	Weighting	Types of Sensors	
		mmWave LD2450 Sensor	LDS LiDAR
Cost of manufacture	0.20	5	5
Durability	0.15	4	3
Complexity	0.20	4	4
Reliability	0.45	5	3
Total score	1.00	4.55	3.40

Table 3.1.1a evaluates sensor technologies for detecting human falls based on manufacturing cost, durability, complexity, and reliability, with high emphasis placed on reliability. The mmWave LD2450 sensor achieved a better score of 4.55 compared to the LDS LiDAR sensor of 3.40. Therefore, the mmWave sensor was chosen due to its reliability, durability, and non-intrusive target tracking.

Table 3.1.1b: Decision Matrix for Alert System for Fall Detection

		Types of Alert System	
Criteria	Weighting	Telegram Bot	SIM800L GSM module
Cost	0.25	5	3
Connectivity	0.15	4	4
Complexity	0.20	4	4
Reliability	0.40	5	5
Total score	1.00	4.25	3.65

Table 3.1.1b compares alert transmission methods using cost, connectivity, complexity, and reliability, prioritizing reliability and cost. The telegram bot provides a cost-effective and highly reliable notification mechanism over network connectivity without requiring external SIM card hardware or cellular plans. This justifies its highest score of 4.25.

Table 3.1.1c: Decision Matrix for Types of LLM Software

		Types of LLM Software		
Criteria	Weighting	Gemini	Claude	Chatgpt
Cost	0.20	5	2	3
Response time	0.20	4	4	4
Performance	0.20	5	5	5
Reliability	0.35	5	5	5
Total score	1.00	4.80	4.75	4.75

Table 3.1.1c compares Large Language Model platforms based on cost, response time, performance, and overall service reliability. While all models offered identical high performance and reliability, Gemini stood out due to its superior cost-efficiency for processing and integration.

Table 3.1.1d: Decision Matrix for Types of Sensors for Smart House

		Types of Sensors	
Criteria	Weighting	IR Sensor	Ultrasonic Sensor
Cost	0.20	4	5
Durability	0.20	5	5
Complexity	0.20	5	5
Reliability	0.40	5	4
Total score	1.00	4.80	4.60

Table 3.1.1d assesses presence sensing options for home automation based on cost, durability, complexity, and heavy weighting on reliability. Although both scored equally in durability and complexity, the IR sensor offered higher operational reliability for smart house applications.

Table 3.1.1e: Decision Matrix for Types of Microcontrollers

Criteria	Weighting	Types of Microcontrollers		
		ESP32 WROOM 32	Raspberry Pi Pico W	Arduino Uno
Connectivity	0.20	4	5	3
Cost	0.25	4	3	5
Durability	0.20	4	4	3
Reliability	0.35	5	5	3
Total score	1.00	4.45	4.15	3.45

Table 3.1.1e evaluates microcontrollers across connectivity, cost, durability, and reliability, heavily prioritizing reliability and cost. The ESP32 WROOM 32 scored the highest with 4.45 as its integrated Wi-Fi/Bluetooth connectivity, low cost, and system reliability make it the optimal main processing unit.

3.2 Overall System Design

Our system is a dementia assistive robot which is going to be a stationary robot aimed to help a person living with early-to-moderate level of dementia illness. Hence, its main purpose is to compensate and help on the specific difficulties that causes trouble for a person with dementia such as forgetting to take their medication, leaving electrical appliances like lights, fan or stove unattended or becoming disoriented at the middle of the night and falling without anyone knowing, so the system main responsibilities are listed in the following table:

Table 3.2: Overall features

Responsibilities	Details
Fall detection and emergency alert	A falling will trigger the emergency branch so that an emergency would never be left unnoticed and urgent help can be reached
Conversational Assistance	User can chat with the robot itself, ask it questions or give commands naturally to help them control something in their home
Memory Support	Robot can remember what the user has done and has not done throughout the day, and tell them something that they cannot recall themselves
Smart Home Automation	Besides controlling the electrical appliances in the house via voice command, unsafe conditions in the house can also be automatically corrected by the system

Our dementia assistive robot generally runs as a single-loop machine, which means that it does one thing at a time and after finishing a task, it returns to the starting/idle state again. Hence, there will be no multitasking or any parallel behaviour for our robot. This means that the robot is always doing exactly one thing at a time, whether it is listening, talking, handling an emergency or waiting, it will never do two of these things at once and idle mode is the only place from which a new decision or

action can be made. We choose to design the system this way as it gives us predictability, as if the robot suddenly malfunction, we can easily trace back step-by-step and there is only one path to inspect instead of an error caused by parallel actions. Furthermore, we also considered about the user's safety, as for instance while in the middle of an emergency alert, it will not do other stuff which would potentially disrupt the flow of the emergency alert and cause potential errors that can be critical. Another thing is the limited hardware since we are only using an ESP32, it has modest processing power and limited memory hence doing things simultaneously would possibly overload it.

In terms of the priority order, we can see from our flow chart below how the logic flows:

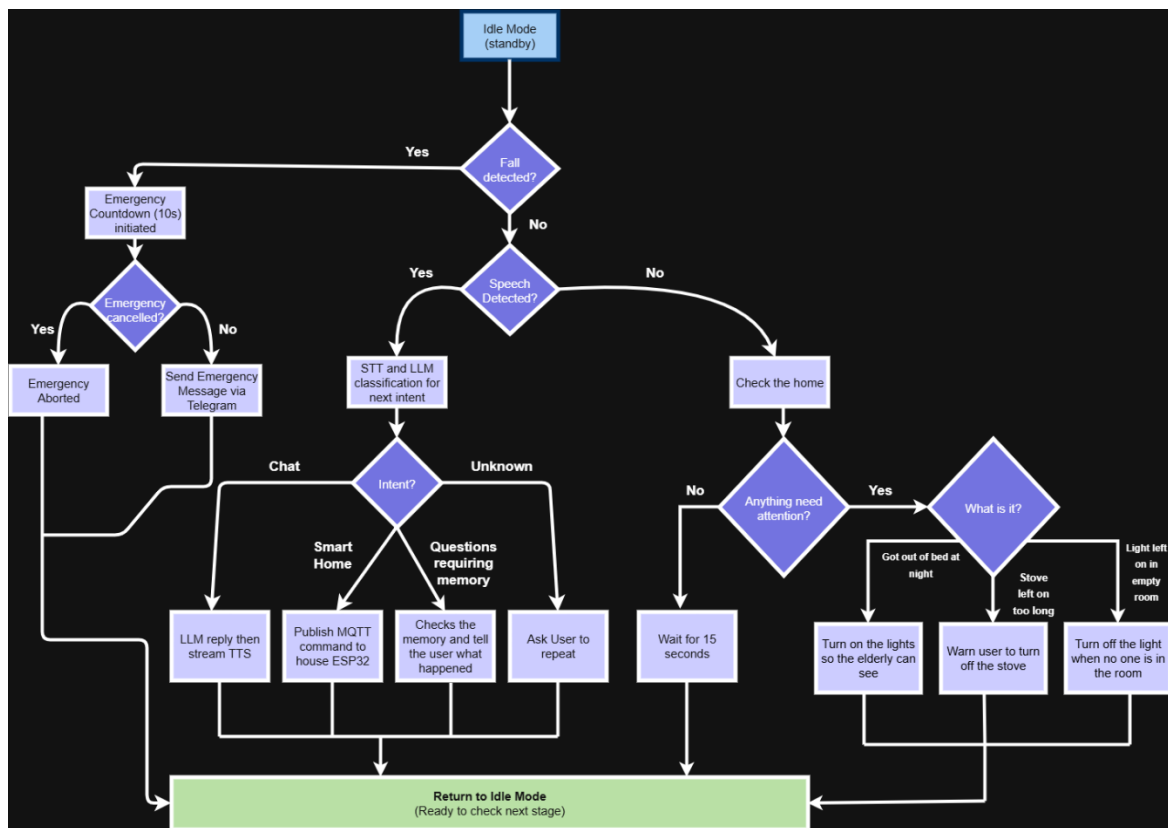


Figure 3.2.1: General Flow Chart of Piper Robot

As seen in Figure 3.2.1, the process begins from idle state. From there, the robot is triggered by either one of the three events in fixed order. We designed it in a way where the most urgent condition is checked first. Firstly, it checks for a fall, as we ranked safety as the number one priority, so if the sensor detects a fall, the robot will immediately enter the emergency branch. Secondly, if no fall is detected, the robot then checks whether the user is trying to talk to it. Thirdly, which is also the default when no fall is detected and no speech is detected, the robot then performs house monitoring and searches through the state of the house and the user's daily routine to check if whether something has been forgotten or looks unsafe.

In our design, there are 3 branches which are emergency, conversation and home monitoring branches.

For the emergency branch, once the sensor detects a fall, the robot will start a 10 second countdown which acts like a grace period just in case of false alarm, so if the emergency is cancelled within that grace period, the alert is aborted and the robot will return back to idle state but if it is not cancelled within the grace period, an emergency message is sent via telegram to the registered caregiver and after finishing this process, the robot will then return to idle state again. The emergency branch is intentionally short and has no dependency on the AI system, hence it will work even if the AI is not working.

On the other hand, for the conversation branch, when speech is detected, the audio signal is passed through speech-to-text (STT) and then the resulting text transcribed will be sent to the Large Language Model (LLM) which processes what the user's intent is and whether it is a basic conversational chat in which the LLM will generate a natural reply which is then converted to speech via text-to-speech (TTS) and streamed back through the speaker on the robot. If the intent is to control something in the smart home, the LLM will produce a structured command which will then be published to MQTT which is directed connected to the second ESP32 installed in the smart home which will perform a specific physical action such as switching on the lights or a fan etc. If the user is asking something that can be recognized by the previous memory stored of the LLM for instance, "have I taken my pills for diabetes today". The robot checks back its memory of the day and tells the user based on the memory recorded what had already happened and so on. If the LLM is unable to classify the user's intent it will ask the user to repeat the speech.

If no fall or speech detected, the robot moves to the final branch which is the home monitoring branch, in which the robot will search and look around the home based on the conditions of who's around, what time it is and what is currently switched on. From the information the robot gathered, the robot will then finalize its decision whether something needs attention. If nothing needs attention, the robot will wait for 15 seconds buffer time and returns to idle state again to begin a new cycle. This buffer allows the robot to come back to Idle state often if nothing else is being done and can look again for falls and speech instead of waiting nonstop, so that it is always paying attention in a way. If something needs attention, the robot will see what it is and try to settle it. For instance, if the

user got out of bed at night, the sensor in their house would detect that and sends a signal to the robot to open the light while the elderly person is walking to prevent any accident from happening. Another situation is, if the stove has been left on for too long, the robot will then send a warning the elderly person to turn it off as a reminder, as this can be catastrophic if left unattended and can be very dangerous which is important as a person with dementia tends to forget things easily. And finally, if a light was left ON in an empty room with no one inside, the robot will also turn it OFF, as the elderly person with dementia might have forgotten to.

3.3 Circuit Design

3.3.1 Circuit Schematic

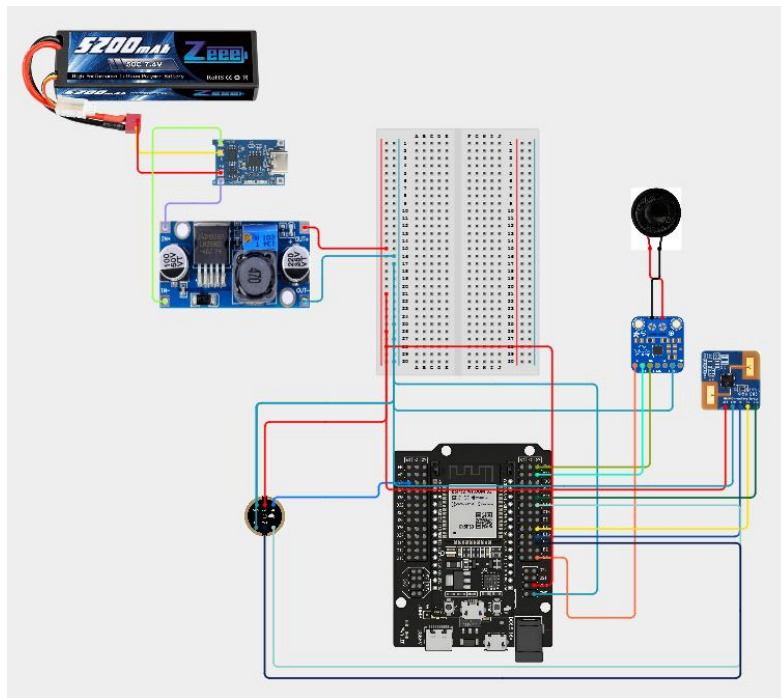


Figure 3.3.1: Electrical Circuit diagram of Dementia Assistive Robot

7.7V with 5.2Ah Lithium Ion Battery is the primary source of energy. It is connected to a charging module and will pass through buck converter to step down the voltage to 5V to produce a clean 5V line voltage source to power up the other components. ESP32 is the main controller in this project. For sensor part, there is a microphone connected to power line and ESP32 for user voice recognition and mmWave Sensor LD2450 for fall detection. An audio amplifier is connected to a speaker for speech from A.I. voice integration.

3.3.2 Component Pin Connections

Table 3.3.2: Pin Connections of each component to ESP32 and Power Source

ESP32 Pin	ESP32 Pin	Component	Signal	Description
VIN (5V)	VIN (5V)	LM2596 Buck Converter	5V Supply	Regulated 5V supply to ESP32 and peripherals
GND	GND	All Components	GND	Common ground connection
3V3	3V3	mmWave Sensor LD2450	3V3	Main Power Source
GND	GND	mmWave Sensor LD2450	GND	Common ground connection
GPIO 16 RX2	D16	mmWave Sensor LD2450	TX	Sensor Transmit, Microcontroller Receive.
GPIO 17 TX2	D17	mmWave Sensor LD2450	RX	Sensor Receive, Microcontroller Transmit.
GPIO 21	D21	mmWave Sensor LD2450	OT2	High/Low output on presence trigger
GPIO19	D19	INMP441 Microphone	SCK (BCLK)	I ² S Bit Clock
GPIO4	D4	INMP441 Microphone	WS (LRCLK)	I ² S Word Select
GPIO39	VN	INMP441 Microphone	SD	I ² S Audio Data Output
ESP32 3.3V	ESP32 3.3V	INMP441 Microphone	VDD	3.3V power supply
GND	GND	INMP441 Microphone	L/R	Common ground connection
GPIO23	D23	MAX98357A Audio Amplifier	DIN	I ² S Audio Data Input
GPIO22	D22	MAX98357A Audio Amplifier	BCLK	I ² S Bit Clock
GPIO15	D15	MAX98357A Audio Amplifier	LRC (WS)	I ² S Left/Right Clock
5V	5V	MAX98357A Audio Amplifier	VIN	5V power supply
Battery (7.4V)	Battery (7.4V)	LM2596 Buck Converter	Input	Battery input
Battery (7.4V)	Battery (7.4V)	Power Switch	Power Control	Main ON/OFF switch

3.3.3 Power Requirement & Calculations

The power required, operating voltage, peak current draw and total current are calculated to find out the minimum and maximum operating runtime of the robot, while preventing power surges and short circuit.

Table 3.3.3: Power requirements of each component and total current draw

Component	Operating Voltage	Peak Current Draw	Total Current
mmWave Sensor (LD2450)	5V	~80 mA	
ESP32-WROOM	5V (via VCC)	~240 mA (Transmit spikes)	

MAX98357A Amp (3W)	5V	~600 mA – 1,000 mA (At high volume)	
INMP441 Mic	3.3V	~1.5 mA	Min 0.922A Max 1.322A

Calculations on how long the robot can last without charging

$$\begin{aligned}
 \text{Max Runtime} &= \frac{\text{Total Battery Capacity}}{\text{Max System Current Draw}} \times \text{Efficiency} \\
 &= \frac{5.2Ah}{1.322A} \times 0.8 \\
 &= 3.15 \text{ Hours (3 hours 9 minutes)}
 \end{aligned}$$

$$\begin{aligned}
 \text{Min Runtime} &= \frac{\text{Total Battery Capacity}}{\text{Min System Current Draw}} \times \text{Efficiency} \\
 &= \frac{5.2Ah}{0.922A} \times 0.8 \\
 &= 4.51 \text{ Hours (4 hours 30 minutes)}
 \end{aligned}$$

3.4 Software and AI Architecture

In terms of the software and AI architecture, the software is split across two different machines which are the ESP32 and a separate computer. The ESP32 handles the physical world applications such as making the sensors work, getting input speech signals from the microphone and sending signals via the speaker. Meanwhile, the computer runs a python backend that handles all the AI processing which is the brain of the system. The ESP32 and the computer are linked together via Wi-Fi over HTTP. We need to design it in such a way as the speech recognition and language models need far more memory and processing power to run which is unable to be achieved if we were to strictly use the ESP32. This also gives us modularity in the future to switch or upgrade our AI models, which we do not have to change anything from the robot, instead we can just change the software part.

3.4.1 ESP32 Communication (Arduino IDE)

The ESP32 overall logic flow and firmware will be written in C++ using Arduino IDE which consists of the following responsibilities:

- Taking readings using the LD2450 mmWave Sensor for the presence, movement and fall of the elderly person
- Recording the speech signals via the microphone and amplifying the recorded audio
- Connecting to the local Wi-Fi network
- Sending the speech recording signals to the backend and receiving a response signal as a reply from the Large Language Model (LLM)
- Playing the received audio through the connected speaker

The logic will be coded in such a way that the firmware will run one task at a time in one single loop and that no AI logic or processing will be done with the ESP32, as its main job is to only detect, transmit, receive and play.

3.4.2 Python Backend (Visual Studio Code)

The backend is basically a Python server that will be developed in VS Code. How it works is that it will expose a small set of HTTP endpoints that the ESP32 initially calls and it will also internally chain all the three steps of the AI pipeline together. The speech signal will first go through Whisper which transcribe our audio to text, then to the Large Language Model (LLM) of Gemini and then we will be using Piper TTS to convert the text generated back to speech. Besides that, it will also write anything worth remembering into the event store and also holds the API keys and publishes the smart-home commands over MQTT.

3.4.3 Whisper Speech-to-Text

For the first part of the AI pipeline, we will be using Whisper which is an OpenAI speech recognition model that run locally on the backend. Its role is to convert our recorded audio sent by the ESP32 into plain text that can be read by the LLM. It was specifically chosen as it handles slow and hesitant speech better than keyword-based recognition software which is especially important for elderly users with dementia as their speech are often unclear and sometimes interrupted or stuttered and that since it is running it locally on the backend, it avoids sending the user's speech online to a third party which helps secure privacy and also remove delays due to network.

3.4.4 Gemini Large Language Model (LLM)

As for our Large Language Model (LLM), we will be using Gemini which is Google's language model. It can be accessed through a cloud API. How it works is that it will receive the transcribed text from the STT and relevant entries information from the robot's memory along with a system prompt that will define the robot's persona for example patient, warm and charming simulating a real person and then with reference to the transcribed text it will process it and return either a spoken reply which will be sent to the TTS, or a smart home command if the user's intent is to control something from the smart home. But overall, Gemini provides good feedback, reasoning and natural conversation that would be perfect for our system.

3.4.5 Piper Text-to-Speech

For the final stage when a response in the form of text is received, we decided to go with Piper for the text to speech conversion which is a super lightweight neural text-to-speech engine that also run

locally. We selected Piper as it is fast enough to run on almost any modest hardware and also it produces a clear, easy to understand human simulated voice that works offline so it does not depend on the internet which cuts down on delays.

3.4.6 Memory and Event Storage

The main problem with an elderly person with dementia is that sometimes they cannot speak or think clearly, so implementing a memory system allowed us to turn the robot from just a chatting assistive device into an actual dementia aid robot, if a person does not remember certain events or what had just happened, the robot can let them know and remembers for them.

The backend will keep a simple diary of the day saved to a small database file on the computer and the robot will write down the important things during that day and each entry is just a time, an event, a value, string value or a brief summary. An example of how the database will look like can be seen in the table below:

Table 3.4.6.a: Example Database for memory storage implementation

Time	Event	Value
08:00	Medicine for diabetes	Taken
09:00	Person	In Kitchen
10:00	Stove	ON
10:15	Stove	OFF
22:30	Person	In Bed

The main things that will be recorded and stored in the database are:

Table 3.4.6.b: List of data category that will be stored in the database

Recorded Category	How so?
Medicine	Recorded when user tells the robot when they have taken their daily meds
Movement	Recorded as user got up and went to bed which will be detected by the mmWave sensor
Appliances	Recorded when the light or stove was turned ON for a long time with no one attending it

Conversations	Short summary of what the robot and the elderly person talked about recently
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This memory system works for example when the user asks a question like “Have I taken my pills for diabetes?”, the robot will look up the diary and with reference to that, the LLM will give an answer.

3.5 AI Voice Interaction

The complete path for our AI voice interaction can be seen in the block diagram below,

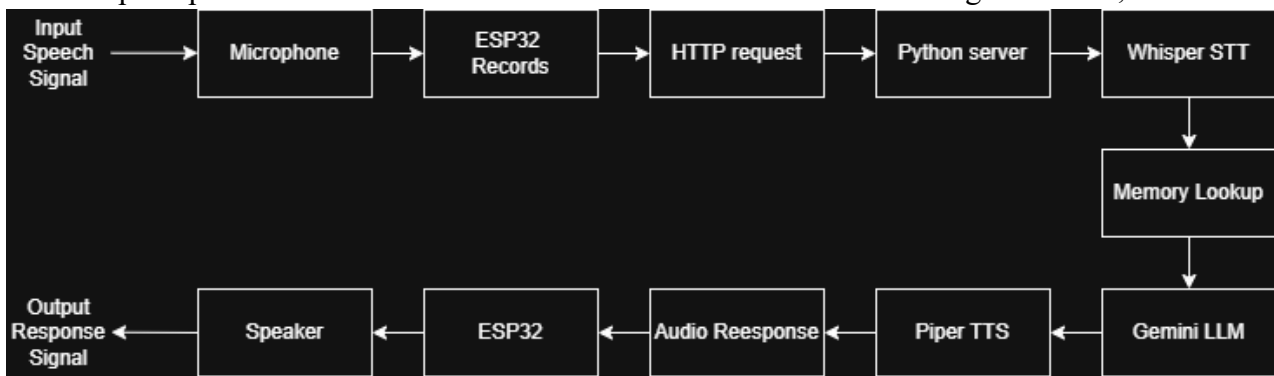


Figure 3.7.1: Block Diagram of AI Voice Interaction

From the block diagram, we can see that our signal constantly change form at different stages, as initially the received input signal is in the form of sound, then it gets digitalized and becomes digital audio. This digitalized audio is then converted into text for processing in which the generated text response is then reconverted into sound signal again.

3.5.1 Speech Input and Speech-to-Text Processing

When the robot is in the “conversational branch”, it will start detecting speech, so when the user speaks, the INMP441 microphone captures the audio signal which will be converted to a digital I²S stream which can be directly read by the ESP32. The ESP32 records the audio which will be sent to a buffer and will stop when the user goes quiet or maximum recording length has been reached. The audio will then be transmitted to Python as a WAV file inside an HTTP POST request over Wi-Fi, in which Whisper will process the audio and match the corresponding sounds to words using the available vocabulary in their database. A resulting output text or a plain string of character will be generated which will be passed on to the next stage.

3.5.2 Large Language Model Processing

The transcribed text received from Whisper will then be sent to the Gemini LLM for processing, but before that the diary is first checked by the backend to find out if the user is asking about something that already happened, hence it will look up by using the “matching entries” method and passes them along with the question asked by the user to Gemini. So, anything found in the diary related will be noted by the LLM itself. Gemini then proceeds to process the information sent by interpreting the meaning of the question or request rather than matching keywords, so that even though the sentence said by the user is grammatically incorrect or has weird sentence structure, the AI will still be able to interpret it. After that, Gemini will generate a short, concise and easily understandable response with a command if user intends to control something from the smart home. The text response will then be passed to TTS and any commands will be sent over to MQTT if the intent is to control smart home.

3.5.3 Text-to-Speech and Audio Output

The text reply generated by Gemini will then be sent to Piper on the same backend machine. Piper then converts the text into speech audio in the form of a WAV file using a natural voice. The audio will then be sent back to the ESP32, in which it will be first amplified by the MAX98357A amplifier and then played out the speaker as sound signal. After completing the whole pipeline, the robot will return to idle mode and start a new cycle and begins to check for fall again.

3.5.4 Communication System of pipeline

How each hardware or software communicates with each other in order to complete the flow of the AI voice interaction pipeline and why is it chosen can be seen in the table below:

Table 3.5.4.a: Communication of the AI pipeline

HTTP Request	ESP32 acts as the body or client and Python is the backend loaded on a computer, so the ESP32 will send a HTTP POST to the server's IP address on the local network. It was chosen as it is simple and well supported by ESP32 libraries and requires no persistent connection to be maintained as well.
--------------	--

Data Format	WAV file with a fixed sample rate and bit depth will be used throughout the pipeline for both conversion for TTS and STT.
Audio Transmission	Due to the fact that ESP32 has limited memory and that the recording is larger than the availability of it, the audio will be streamed in chunks rather than the whole thing in which the chunks will be reassembled by the server before passing to Whisper.
Flow State	The agents for the whole pipeline whether is it the ESP32 or backend will be doing one thing at a time and strictly following a single path flow, for instance while Piper is converting the file to WAV audio, the ESP32 will wait for the HTTP response instead of listening to another request again from the user. This ensures that everything is synchronised and error can be reduced.
Error Handling	In the case of errors, the communication system designed by us will handle the situation accordingly: • No Wi-Fi – plays a stored local message telling the user it cannot reach the AI • No speech detected or Whisper returns nothing – backend replies asking user to repeat sentence • Corrupted or empty audio - backend replies asking user to repeat sentence • Nothing found in the diary – the robot will generate a response saying it does not have a record of it

3.6 Fall Detection and Emergency Alert System

Fall detection is used as the emergency response to alert caregivers or family members through Telegram notification. Fall is detected using a 24GHz mmwave radar sensor (LD2450) through human presence, 2-D position coordinates, and velocity. The change in position and velocity is calculated and compared against the predefined threshold value, which will be determined experimentally by recording the radar sensor's data during normal activities. A sudden or large change in these parameters triggers a 10s confirmation interval. If the movement remains above the threshold value for 10s, the algorithm classifies it as a fall and immediately sends an emergency alert message through Telegram.

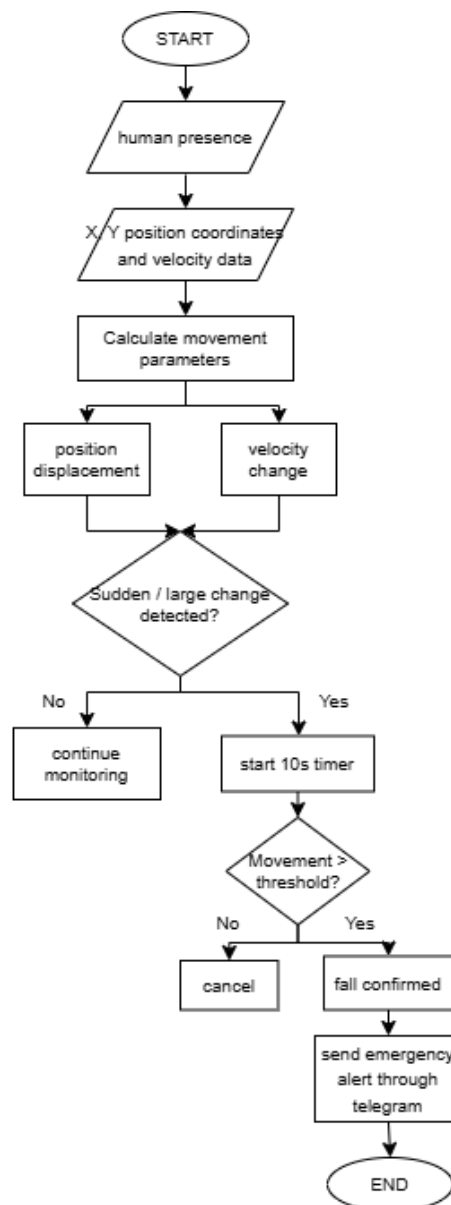


Figure 3.8: Flowchart for fall detection and emergency alert

3.7 Tools, Materials and Software

Hardware Components:

Table 3.7.a: Hardware components list and its role

Item	Role
ESP32 (on the robot)	Main Microcontroller that reads the mmWave sensor over UART, capture and plays audio over the devices and communicates with the Python backends
ESP32 (Smart House)	Receives MQTT commands and control the house's electrical appliances
LD2450 mmWave Sensor	Detects human presence, movement, distance and fall-like motion, but mainly used in the fall detection mechanism
INMP441 MEMS Microphone (I ² S)	Records the user's speech
MAX98357A Audio Amplifier (I ² S)	Converts the I ² S audio stream from the ESP32 into an amplified analogue signal
Speaker (4Ω/ 3W)	Outputs the voice response
LM2596 Buck Converter	Steps the 7.4 V battery down to a regulated 5 V rail to supply the ESP32 and the audio amplifier
7.4 V Lithium-Ion Battery Pack	Primary power source to provide voltage and current
Power Switch	ON/OFF control for the system
Breadboard, Jumper Wires, Headers	Provides stable connection for all the electrical components during prototyping

Software and Development Tools :

Table 3.7.b: Software and development tools list and its role

Item	Role
Arduino IDE	Writing and compiling the code for the ESP32 firmware
Visual Studio Code with Python	Development environment for the Python backend that handles the entire AI pipeline
Flask	Provides the HTTP endpoints that the ESP32 calls to send audio and sensor events and to receive reply
Whisper	Convert recorded audio into text
Google Gemini	Large Language model which interprets user's intent and generates a response according to what the user wants
Piper TTS	Converts the generated text reply to natural sounding audio

SQLite	Local database acting as a memory to store logs and events or conversational data
MQQT Broker	Relays smart home commands to the house ESP32 and acts like a middleman
Telegram Bot API	Sends emergency when emergency alert is triggered to the caregiver
PubSub Client/ ArduinoJson / I ² S libraries	ESP32 side libraries for firmware (MQQT messaging, Json encoding and audio handling)
Github and Git	Version control and code sharing between teammates

Datasets:

Table 3.7.c: Datasets used for the systems

Item	Role
Whisper Vocabulary dataset for Speech-to-Text	Ready-trained and no additional speech data is required to train the model
Piper voice model conversion for Text-to-Speech	Provides the robot's synthesis voice and speech to voice conversion algorithm
Memory or logs of events that the user has done	Activities, conversations and other daily routine data collected during the day and used for reminders and automation rules

3.8 3D Design

The physical structure of PIPER was modelled using a computer-aided design (CAD) software, Autodesk Fusion 360, prioritising operational stability and elder-friendly ergonomics. The primary objective of the mechanical architecture is to provide a durable, lightweight enclosure that houses complex electronics and active hazard-clearing hardware while maintaining an approachable form factor for elderly users.

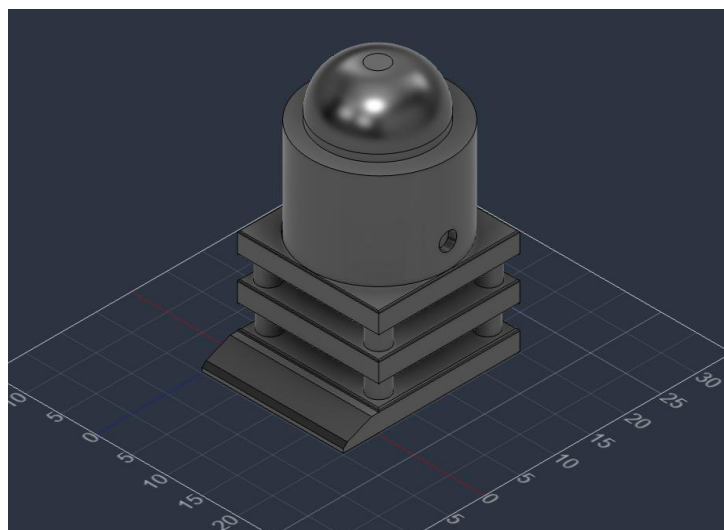


Figure 3.8 Final 3D Design

3.8.1 Design Considerations

There were multiple aspects of design that were taken into consideration before finalising the product. Those being, safety and approachability, low centre of gravity, and finally optimal range of view.

1. Safety and approachability: The outer shell features rounded geometries and soft edges, minimizing sharp corners that could present hazards or evoke an intimidating, appearance.
2. Low centre of gravity: The base of the robot features a multi-tiered base that allows the physical weight distribution to be heavily biased on it. The lowest level of the base is where the main heavy components such as battery packs, and the main circuitry are stored to ensure a steady and stable base to ensure the balance of the robot. This low centre of mass prevents tip-overs when any sudden movements are inflicted upon it. Apart from that, the reason the main circuitry is placed at the bottom is to ensure that the wires within the robot don't get jumbled up as compared to if the circuitry was placed directly into the body of the robot.
3. Optimal range of view: the base is a multi-tiered base to store majority of the components. Therefore, the top layer of the base is where the sensors will be placed with an open gap all around it to ensure a 360° range of view.

3.8.2 Material selection

To align with modern eco-friendly design standards and facilitate rapid prototyping, PIPER's structural frame and exterior panels were manufactured using PolyLactic Acid (PLA). The main reason for this selection is that PLA is a corn-starch derived, biodegradable filament that offers a low carbon footprint without requiring high-energy production environments. The production of raw PLA polymer generates roughly 68% fewer greenhouse gas emissions and consumes up to 65% less embodied energy compared to petroleum-derived plastics.

3.9 Safety and Ethical Considerations

Designing, fabricating, and experimentally validating a mobile companion robot introduces physical, electrical, and operational hazards. PIPER is powered by high-capacity Lithium-Polymer (LiPo) / Lithium-Ion battery packs capable of high discharge currents. Electrical short-circuiting, overcharging, or physical puncture poses risks of thermal runaway, fire, or chemical leakage. Mitigation measures to be taken are the power distribution board incorporates inline fast-blow fuses and solid-state overcurrent protection. Batteries are charged using smart balancing chargers inside fire-retardant LiPo safety bags.

Deploying an AI-driven robot within the private living spaces of elderly individuals requires strict adherence to ethical design principles. PIPER addresses key ethical dimensions surrounding data privacy, algorithmic fairness, and human autonomy. Elderly users, particularly those experiencing early-stage cognitive decline, may not fully grasp how a companion robot functions, leading to over-

reliance or fear. Mitigation measurements to be taken are experimental trials involving human participants operate under a formal Informed Consent framework, offering clear, plain-language explanations of PIPER's capabilities and limits. The robot incorporates visible status indicators to communicate when it is actively listening, ensuring full operational transparency.

3.10 Sustainable Design Practices

To practice sustainable engineering in robotics, it requires looking beyond immediate operating efficiency to evaluate environmental, physical, and financial impacts across the system's entire lifecycle. From raw material extraction to final end-of-life disposal.

Carbon footprint reduction is one of the main priorities in this project. Hence, PIPER's physical construction prioritizes low-emission materials and optimized printing techniques. Firstly, is to select a material with the lowest carbon emissions. Raw PLA generates 2.0 kg CO₂ e/kg, cutting manufacturing carbon emissions by approximately 55% compared to traditional ABS (4.5 kg CO₂ e/kg) and aluminium (8-12 kg CO₂ e/kg). Apart from that, PLA prints at lower nozzle temperatures (200°C), reducing electricity consumption during fabrication compared to high-temp plastics.

Apart from material selection, that would be strategies to optimise energy efficiency. Running lightened AI models directly on the robot consumes less than 15W under heavy load, avoiding power-heavy cloud data transfers. Having an automated idle state when no user activity is detected, Finally, PIPER is designed to be disassembled when it is no longer usable to the user. Components are planned to be connected using screws and to have an attachable and detachable body that can be taken apart easily. Moreover, waste PLA parts can be industrially composted into harmless CO₂ and water or shredded and re-extruded into recycled 3D printer filament. As for the electrical components, sensors, microcontrollers, and cameras use modular pin connectors, allowing them to be harvested for future hardware projects. Battery packs sit in removable trays for easy e-waste recycling.

Whole-life cost analysis (WLC):

To ensure economic feasibility and long-term sustainability of PIPER, a 5-Year Whole-Life Cost Analysis is conducted as the robot is expected to last 5 years. Based on the Bill of Materials and energy consumption calculations, the whole life cost can be estimated.

Table 3.10: Whole-life cost analysis

Cost category	Description	5-year cost (RM)
Initial BOM	Robot components	234.90
Electricity Consumption	Energy 4 hours active + 20 hours standby per day	22.90
Battery Replacement	1× Li-ion pack replacement (5.2 Ah, cycle life degradation at ~2.5 yrs)	29.00
Maintenance & Repairs	Minor PLA, wiring and mechanical maintenance	15.00
End-of-Life Disposal	Battery, electronics and PLA disposal/recycling	10.00
Total 5 Year WLC		311.80

From Table 3.10, even though the robot initially costs a total of RM234.90 to build, there are additional costs during its lifetime such as electricity, battery replacement, maintenance and repairs, and even end-of-life disposal. This shows that Piper has a relatively low lifetime operating cost compared with its initial construction cost. The initial BOM contributes approximately 75% of the estimated 5-year WLC, while energy, battery replacement, maintenance, and disposal together contribute the remaining 25%.

4 Project Plan

This section outlines the project management framework to execute the project within the 14-week timeframe through defining the work breakdown structure, Gantt chart, expected milestones, bill of materials, risks and mitigation.

4.1 Task Breakdown (Work Breakdown Structure)

This project is broken down into smaller manageable tasks in the work breakdown structure in Figure 4.1.

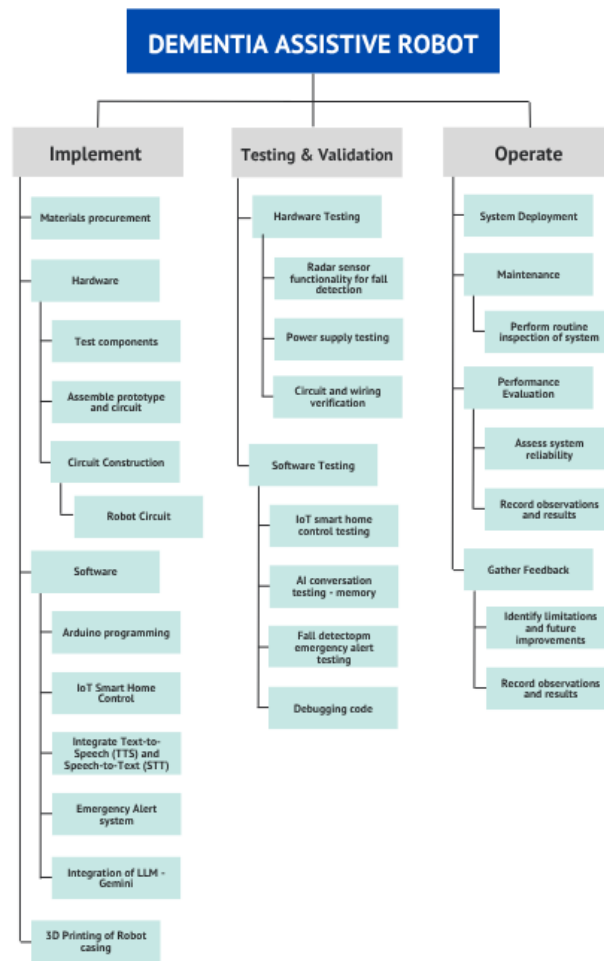


Figure 4.1: Work Breakdown Structure

4.2 Gantt Chart

Figure 4.2 shows our complete project's timeline from implementation to operate phase over 14 weeks. Most tasks run in parallel for efficiency while some run sequentially with dependencies represented using arrows that connect from one task to another. The key milestones are marked using a diamond symbol.

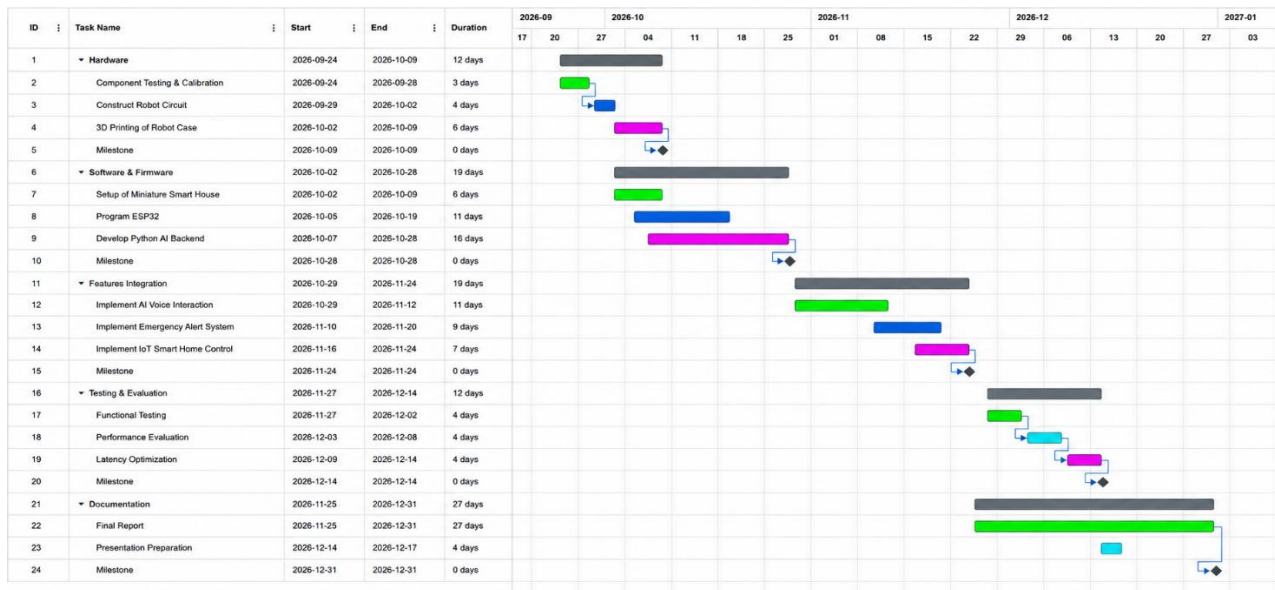


Figure 4.2: Gantt Chart

4.3 Expected Milestones

The key milestones of our project are listed in table 4.3 with the expected completion dates.

Table 4.3: Expected milestones and dates

No.	Milestones	Expected Completion Date
1	Hardware assembly of dementia assistive robot	9 October 2026
2	Software development ESP32 and Python AI Backend	28 October 2026
3	Feature Integration of robot features	24 November 2026
4	System testing and performance evaluation	14 December 2026
5	Final project handover and documentation	31 December 2026

4.4 Bill of Materials

Table 4.4 lists the estimated budget of our project totalling up to RM234.90, which lies within the allocated budget of RM500. This leaves us with a buffer of RM265.10 (approximately 53% of total budget) for emergency expenses, spare components for testing or price fluctuations.

Table 4.4: Estimated bill of materials

Unit	Component	Quantity	Cost / Unit (RM)	Total (RM)
1. Main Control Unit	ESP32-WROOM-32	1	29.90	29.90
	Micro USB Cable	1	2.00	2.00
2. AI Voice Interaction	INMP441 I ² S MEMS Microphone	1	11.88	11.88
	MAX98357A I ² S Audio Amplifier	1	5.85	5.85
	3 W 40 mm 4 Ω Full-Range Speaker	1	9.68	9.68

3. Fall detection & alert system	mmWave Sensor LD2450	1	52.75	52.75
	Active buzzer	1	1.50	1.50
4. Power Supply	7.4 V Li-ion Battery Pack (2200–5000 mAh)	1	29.00	29.00
	LM2596 Buck Converter	1	4.45	4.45
	Power Switch	1	2.50	2.50
5. Mechanical body	3D Printing casing and body	1	74.49	74.49
6. Wiring & Assembly Unit	Soldering board	1	4.90	4.90
	Wires & Jumper Cables	Roll/multiple	1.50	6.00
TOTAL				234.90

The highest cost arises from the 3D printed mechanical body, followed by the ESP32 microcontroller and power supply which are the core components for the system to run.

4.5 Risks and Mitigation

The limitations, risk, mitigation and future improvements of the project are identified in Table 4.5.

Table 4.5: Limitations, risk, mitigation and future improvements

Limitation	Risk	Mitigation	Future Improvement
ESP32 has limited computational capability	AI processing may be too demanding for the ESP32	Offload AI processing to the Python backend	Use a more powerful processor if local AI processing is required
AI depends on network communication	High latency or network failure may delay or interrupt responses	Measure latency and optimise the communication pipeline	Improve local processing / add offline fallback functions
Limited battery power	Robot may not operate for the required duration	Perform power-budget and battery-duration testing	Use a higher-capacity or more efficient battery
Fall detection may be inaccurate	False positives or missed emergency events	Conduct controlled testing and tune detection criteria	Improve sensing and detection algorithms
Microphone sensitivity may be affected by environmental noise and speaking distance	Speech may not be captured clearly, resulting in inaccurate STT or missed voice commands	Position the microphone appropriately	Use a higher-sensitivity microphone or a microphone with noise cancellation

Prototype hardware is limited by budget	Unexpected component costs may exceed the available budget	Maintain the remaining budget as contingency	Reduce component costs or optimise the design for future versions
Voice interaction requires multiple processing	Longer response time may reduce user experience	Measure STT, LLM, TTS latency separately	Optimise each stage and use local processing

5 Conclusion

This project presented the complete design and implementation plan for PIPER, an integrated dementia assistive robot developed to address the growing public health challenges of an aging population. PIPER unifies real-time emergency fall detection, natural AI voice interaction, and daily memory assistance into a single, cohesive platform designed specifically for homebound elderly individuals suffering from early-to-moderate dementia.

Throughout IDP 1, the main accomplishments made were system architecture and hardware selection, AI and software pipeline, integration of safety feature, and sustainable ergonomic design. The system architecture standardises on an ESP32 microcontroller paired with an onboard 24GHz mmWave radar sensor (LD2450) for non-intrusive, privacy-preserving fall detection, alongside high-fidelity audio peripherals (INMP441 microphone and MAX98357A amplifier) for voice processing.

To enable seamless user interaction, a robust multi-stage AI and software pipeline was designed using local Whisper Speech-to-Text (STT), Google's Gemini Large Language Model (LLM) integrated with a custom event memory database, and local Piper Text-to-Speech (TTS) to deliver natural, context-aware human-robot engagement targeting an average response time under 3 seconds. For user protection, the safety and smart home integration features an automated Telegram alert mechanism capable of notifying caregivers within 10 seconds of a confirmed fall. Finally, the sustainable ergonomic design establishes a 3D-printable PLA robot enclosure engineered with a low centre of gravity and rounded geometry, effectively prioritizing user safety, eco-friendly manufacturing, and cost efficiency.

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Appendix

None.